IDENTIFICATION OF RISK - FINANCIAL VS. NON-FINANCIAL RISK

Learning Outcome

- identify financial and non-financial sources of risk and describe how they may interact

Having laid the framework for understanding the concept of risk management and risk governance, we now move into the implementation of the process. One of the first important parts of the process is the identification of risks. In this reading, we identify two general categorizations of risks. The first is the set of risks that originate from the financial markets. Accordingly, we refer to this type of risk as **financial risks**. The second group of risks includes those that emanate from outside the financial markets. As such, we refer to these as **non-financial risks**. Although most risks ultimately have monetary consequences, we reserve the term "financial risks" to refer to the risks that arise from events occurring in the financial markets, such as changes in prices or interest rates.[11] In this reading, we will consider the types of financial and non-financial risks faced by organizations and individuals.

Financial Risks

The risk management industry has come to classify three types of risks as primarily financial in nature. The three primary types of financial risks are market risk, credit risk, and liquidity risk. **Market risk** is the risk that arises from movements in interest rates, stock prices, exchange rates, and commodity prices. This categorization is not to say that these four main factors are the underlying drivers of market risks. Market risks typically arise from certain fundamental economic conditions or events in the economy or industry or developments in specific companies. These are the underlying risk drivers, which we will cover later.

Market risks are among the most obvious and visible risks faced by most organizations and many individuals. The financial markets receive considerable attention in the media, and information on financial market activity is abundant. Institutional investors and many corporations devote considerable resources to processing this information with the objective of optimizing performance. Many individuals also devote considerable attention to market risk, and financial publications and television and radio shows are widely followed in the general population. The state of knowledge in risk management is probably greatest in the area of market risk.

The second primary financial risk is credit risk. **Credit risk** is the risk of loss if one party fails to pay an amount owed on an obligation, such as a bond, loan, or derivative, to another party. In a loan, only one party owes money to the other. In some types of derivatives, only one party owes money to the other, and in other types of derivatives, either party can owe the other. This type of risk is also sometimes called default risk and sometimes counterparty risk. As with market risk, the root source of the risk can arise from fundamental conditions in the economy, industry, or weakness in the market for a company's products. Ultimately, default is an asset-specific risk. Bond and derivatives investors must consider credit risk as one of their primary decision tools.[12] Similar to market risk, credit risk is also a highly visible risk with considerable attention paid to defaults, bankruptcies, and the stresses arising from inadequate cash flow in relation to leverage. Credit risk is a particularly significant risk in that although market prices can go down and bounce back up, defaults and bankruptcies have extremely long-term implications for borrowers.

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unknowingly. This third risk is **liquidity risk**, which is the risk of a significant downward valuation adjustment when selling a financial asset. In order to sell an asset, a party may need to reduce the price to a level that is less than the marked value or the seller's assessment of the asset's true value based on the fundamentals of the asset. In certain market conditions, the seller must make a significant price concession. Having to make price concessions is not necessarily unusual and does not imply a poorly functioning market. Indeed, given no shift in demand, a rightward shift of a supply curve in order to sell a larger quantity is entirely consistent with the notion that a seller must lower the price to sell a greater quantity.

All assets have transaction costs in the market, such as the bid–ask spread. The existence of a sell price that is less than a buy price, however, is not a risk but simply a cost. It is the *uncertainty* of that valuation spread that creates this type of risk. Thus, liquidity risk could also be called "transaction cost risk." The liquidity risk of a $10 stock purchased for $10 is not the risk that one would receive the "bid" price of only $9.99 right after one bought it. That $0.01 spread is a known cost when the stock is purchased, so it is not a risk. The risk is that this spread cost might increase dramatically as a result of either changing market conditions or attempting to maintain a position significantly larger than the normal trading volume for the stock. This problem becomes a serious issue for risk management when the liquidation price falls to less than the seller's estimate of the fundamental value of the asset. Although this risk is often associated with illiquid assets,[13] it really stems from a couple of sources. First, market liquidity varies over time and the market for specific assets may become less liquid; second, as the size of a position increases, the cost and uncertainty associated with liquidating it will increase. In some extreme cases, there may be no price above zero at which the seller can sell the asset.

Of course, one might argue that the cost of illiquidity, and liquidity risk, should thus be part of the investor's assessment of fundamental value, and indeed it is for many analysts. If not, liquidity risk can sometimes be confused with a form of valuation denial in which investors believe that they paid an appropriate price and that the market has not converged to its true value. But less liquidity means a thin market and a lack of investor interest, which may be fertile ground for investment opportunities. Although lack of liquidity can offer benefits, such as the opportunity to buy an asset well before everyone else sees that it is an attractive investment, liquidity risk is generally considered to be a negative factor with which risk managers and indeed all investors must contend.

Non-Financial Risks

Recall that we refer to financial risks as those arising primarily from events occurring in the financial markets. Although most risks have monetary consequences, there are a number of risks that are typically classified as non-financial in nature. These risks arise from a variety of sources, such as from actions within the organization or from external origins, such as the environment as well as from the relationship between the organization and counterparties, regulators, governments, suppliers, and customers.

One important risk of this type is closely related to default risk but deals more with the settling of payments that occur just before a default. This risk is called settlement risk. As an example, suppose Party A enters into a forward contract to purchase ¥200 million of Japanese government bonds from Party B. At expiration if all goes well, Party A would wire the money and Party B would transfer the bonds. Each party fulfills its obligation expecting that the other will do so as well. However, suppose Party A wires the money but Party B does not send the bonds because it has declared bankruptcy. At this point, Party A cannot get the money back, except possibly much later through the potentially slow and cumbersome bankruptcy process.[14] Although the financial consequences are very high, the root source of this risk is the timing of the payment process itself.

Organizations face two types of risks related to the law, and as such, this risk is referred to as legal risk. One risk is simply the risk of being sued over a transaction or for that matter, anything an organization does or fails to do. In financial risk management, however, the

corresponding gain to Bank E. But suppose that Party F then identifies a legal issue that it interprets as giving it the right to refuse to pay. If the court upholds Party F's position, Bank E could incur a loss. Litigation always involves uncertainty because even a seemingly weak case can prevail in court.

The following three non-financial risks are related: regulatory risk, accounting risk, and tax risk. They could even be collectively referred to as compliance risk because they all deal with the matter of conforming to policies, laws, rules, and regulations as set forth by governments and authoritative bodies, such as accounting governing boards. Obviously the regulatory, accounting, and tax environment is always subject to change, but the rapid expansion of financial products and strategies in relation to the relatively slow manner in which government and private regulators are able to respond means that laws and regulations are nearly always catching up with the financial world. When these laws and regulations are updated, it can result in significant unexpected costs, back taxes, financial restatements, and penalties.

Another type of non-financial risk is model risk, which is the risk of a valuation error from improperly using a model. This risk arises when an organization uses the wrong model or uses the right model incorrectly. A simple example applicable to both a portfolio manager and a corporate analyst is the assumption of constant dividend growth in the dividend discount model when, in fact, growth is not constant.

Closely related to model risk is tail risk—more events in the tail of the distribution than would be expected by probability models. This risk is a facet of market risk, but it also infects valuations and models when it is ignored or mishandled. Tail risk is known to be especially severe for the normal distribution, which tends to be overused in modeling. As an example, consider the monthly returns on the S&P 500 Index from January 1950 to October 2018. The monthly average return was 0.70%, and the monthly standard deviation was 4.10%. If we rank the monthly returns, we would find that the largest negative return was −21.76%, which occurred in the well-known market crash of October 1987. With a normal distribution, we would find that a return that low would occur only once every 2,199,935 years.[15] The second and third worst monthly returns of −16.94% (October 2008) and −14.58% (August 1998) would occur only once every 6,916 and 654 years respectively. If the normal distribution is a realistic descriptor of returns, results of these magnitudes should *never* have occurred in recorded market history, and yet we have seen three such instances. Interestingly, according to the normal distribution, the largest positive return of 16.30% in October 1974 would occur only once every 888 years. Technically, one could argue that if we go another 2,199,935 years and do not observe a monthly return as low as −21.76%, then the assumption of a normal distribution might seem reasonable, but it seems safe to reject the normal distribution for at least another two million years. Similar comments can apply to the second and third worst returns albeit over shorter periods.

Many quantitative models (e.g., option models) and decision models (e.g., portfolio construction and asset allocation, relying on variances and covariances in analysis and decisions) ignore the existence of fat tails in returns; as a result, market risk is often considered and dealt with in an oversimplified fashion. Tail risk, as the term is used in practice, is important and is discussed separately because financial professionals realize the implicit failure of modeling market risk. More plainly, ignoring tail risk is a form of model risk. And although tail risk might seem more of a financial risk than a non-financial risk, the mistake occurs internally, arising from poor choices made in modeling.

Most of the internal risks faced by an organization are often grouped together and referred to as operational risk. **Operational risk** is the risk that arises from inadequate or failed people, systems, and internal policies, procedures, and processes, as well as from external events that are beyond the control of the organization but that affect its operations. Although the factors that give rise to such risks can arise externally, the risks themselves are largely internal to an organization because it would be expected to have its

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aware of the vulnerability to employee theft, given the ease with which so many employees have access to accounts and systems for making entries. But even perfectly honest employees make mistakes, and some can be quite costly. The employee who credits someone's account $100,000 for a $100 deposit may have made an honest mistake, but it is a mistake that could quickly lead to the rapid disappearance of money. In the past, employees up to senior management have been guilty of perpetrating accounting fraud, not necessarily for their own direct benefit but to make the company look better.

In banks and other companies that trade in the financial markets, there is the risk that a trader or portfolio manager will fail to follow laws, rules, or guidelines and put the company at great financial risk. This individual is commonly described as a "rogue trader." Personified by Nick Leeson of Barings Bank, who in 1995 destroyed the 200-year old company by engaging in a series of highly speculative trades to cover up losses, the rogue trader has become a standard concern of risk managers. Although it was never clear if Leeson's trades were truly unauthorized, his legacy left the fear that institutions bear the risk that one trader can imperil the entire organization by making large and highly speculative trades that put the bank's entire capital base at risk. In essence, a rogue trader is a trader who engages in risky transactions without regard for the organization's limits or conforming to its controls.

Organizations are also threatened by business interruptions, such as those caused by extreme weather and natural disasters. Events such as floods, earthquakes, or hurricanes can cause significant damage and temporarily shut down an organization. Although extreme weather and natural disasters are external forces that are completely out of the control of an organization, it does not excuse the organization from having the appropriate internal procedures for managing problems caused by their external environment. Simple and fairly low-cost actions, such as having generators, backup facilities, or providing employees the option to work remotely, can go a long way toward keeping employees working during extreme weather events and when natural disasters strike. Yet, some organization have not heeded inclement weather forecasts. Failing to react to warnings can result in considerable loss.

In a world that is increasingly digital, cyber risk is a major operational risk that spares no organization and that can have significant consequences. Organizations are expected to understand and manage the risk associated with the disruption of or failure related to their information technology (IT) systems. For example, a hacker breaking into a company's IT system and stealing customer or client data is an external threat. Hacking, however, is not simply a random act of mischief. Companies are aware of the threat of hackers, and hackers can break in to a system only if that system is vulnerable. An organization is responsible for ensuring cyber security and establishing sufficiently robust IT safeguards, such as data encryption, to deter hackers from breaking in and either stealing or causing disruption. Cyber-attacks and data breaches can have serious reputational and compliance consequences. For example, all organizations targeting European citizens must comply with the General Data Protection Regulation (GDPR) and notify regulators and data subjects of any data breaches regarding sensitive personal information within 72 hours. Failure to do so can lead to fines of several million euros, including for organizations based outside the European Union. In addition to the threats posed by hackers and viruses, even secure IT systems themselves are a particular source of risk. Programming errors and bugs can create the possibility of costly mistakes.

Terrorism is another form of operational risks that poses a threat to organizations and individuals. The 1993 attacks on the World Trade Center led many companies to recognize that the New York City financial district was a major terrorist target and that, as such, their operations could be shut down by these acts of violence. When the more destructive attacks of 11 September 2001 occurred, many organizations had already established backup facilities sufficiently far away from that area. Of course, such risk is not

discuss insurance later, but most companies would much prefer to take proactive steps toward prevention than to incur the inconvenience of losses and then have an outside organization compensate them for their losses.

Solvency risk is the risk that the organization does not survive or succeed because it runs out of cash, even though it might otherwise be solvent.[16] This was probably the most underappreciated component of risk prior to the financial crisis of 2008.[17] The collapse of Lehman Brothers was often associated with an excess of leverage, which was certainly a key factor in its failure. But it was solvency risk that forced the company into bankruptcy. Almost overnight, Lehman's liquidity disappeared because most funding sources would no longer willingly bear Lehman's counterparty risk. Even if it had experienced large market gains on the day it went under, it had already been destroyed by solvency risk. Across the entire financial industry, from hedge funds to pension funds, painful but valuable lessons were learned about the critical importance of funds availability and solvency risk, even if all other risks were well-aligned. Solvency risk is now viewed as one of the key factors in running a successful hedge fund because investors are extremely sensitive to not recovering their investment in the event of a "run on the fund."

Solvency, in the personal or institutional sense, is the availability of funding to continue to operate without liquidating—or at a less extreme level, to be able to make good on liabilities and meet one's cash flow requirements. Solvency risk is the ultimate example of the importance of taking an enterprise view of risk management. For example, a university's investment officer might have a perfectly well-balanced set of risks in the endowment portfolio when viewed in isolation. But as a part of a university, the portfolio may be affected by a deep recession because the university's professional degree revenue, grant money, and donations will fall at the same time as the portfolio's investment value and cash distributions are in decline. Although the endowment and university may survive, it might be necessary for the endowment to take many emergency actions that impair its value, simply attributable to the overall solvency risk and the ultimate need of the enterprise to not run out of cash.

Solvency risk is easily mitigated, though never eliminated, by a large number of possible safeguards, none of which is free. Many businesses produce short-term higher returns by essentially ignoring solvency risk, but in doing so, they are not managing risk very well. Since the 2008 crisis, most businesses are keenly aware of the consequences of bad solvency management, and have taken such steps as using less leverage, securing more stable sources of financing, investing in models to provide more transparency on solvency risk, incorporating solvency risk at an enterprise level in risk governance, and holding more cash equivalents and assets with less liquidity risk.

Individuals can also face a number of risks of an operational nature. These include hackers breaking into one's computer and the threat of burglary and robbery. One of the most commonly cited risks for individuals is identity theft. For individuals, however, we consider their primary non-financial risks to be related to their life and health as well as other life-changing events.

Obviously, the health of an individual is an extremely important risk. Poor health can result from poor choices in life, but it can also arise from factors that are outside the control of the individual. These risks can result in direct health care expenses, reduced income because of disability, and reduced lifespan or quality of life. People vary widely in the risk management strategies they undertake to control their health, such as in their choices in diet, exercise, preventive health care, and avoidance of undue health risks. Some individuals address only their financial exposure to health risks, and still others do not take proactive steps to address this risk at all.

Closely related to health risk is mortality risk—the risk of dying relatively young—and longevity risk—the risk of outliving one's financial resources. Not only are these risks a primary determinant of the quality of life, they are also critical factors in investment

need only know the group average mortality. Mortality tables are reasonably accurate, so these institutions have relatively precise estimates of death rates for groups as a whole. Individuals themselves, however, clearly do not know how long they will live. People who use defined contribution plans must therefore build portfolios and control retirement distributions so that their assets outlive them, which is difficult to do when they do not know when they will die. No one wants to outlive their money, but with an increasingly aging population and good health care, this problem is becoming a greater concern.

There are a number of other major non-financial risks that individuals face, which are generally involved with some sort of life-changing disaster. The largest ones—fire, natural disaster, or massive liability stemming from harming others, such as in a car accident—are generally considered “property and casualty” risks and are insured as such.

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